Final Report on the Emotion Classifier Project

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Abstract

Facial Expression Recognition (FER) is heavily constrained by data scarcity, severe class imbalance, and inherent label ambiguity. This report systematically evaluates multiple deep learning architectures to establish an optimal pipeline for emotion classification. We rigorously benchmark models including EfficientNetV2-S, Vision Mamba (Vim), and ResNet18 augmented with Squeeze-and-Excitation (SE) blocks. Furthermore, we investigate the efficacy of transfer learning utilizing an InsightFace ResNet-50 backbone, extensive data augmentation, and advanced anti-bias loss functions. Our empirical results demonstrate that ResNet18+SE achieves the highest overall accuracy, outperforming alternative architectures by approximately 2 percent. Additionally, we evaluate model interpretability, concluding that Integrated Gradients yields significantly more precise visual explanations for low-resolution facial inputs than standard Grad-CAM. The findings highlight the critical importance of architectural selection and the limitations of forcing mathematical fairness on noisy datasets.

1. Introduction

How do humans communicate?

Most people would immediately think of spoken or written language. Yet communication goes beyond words. An essential component of human interaction is *emotion*. Emotions form a key part of nonverbal communication and have been widely studied within the field of psychology.

1.1. What is facial emotion recognition

Facial emotion recognition (FER) refers to the ability to identify human emotions computationally based on images. Other types of emotion recognition tasks approach the same problem by analyzing speech or writing to reconstruct a person's emotional state. As past research has shown, understanding human emotions is no trivial task for an algorithm. The best results in the field of FER have been achieved using complex model architectures and significant amounts of training data. However, the results are quite promising. Recent publications have exceeded accuracies of 90% [12] in their studies.

1.2. Why is FER important

Given that humans themselves are capable of emotion recognition, it may not be immediately clear why research in this field is so important. The answer to this question is that characterizing a person's feelings is a key part of many important sectors such as [24]:

- Healthcare
- Public safety
- Crime detection
- Advertising

By automating the emotion recognition process needed in these areas, the risk of human error can be significantly reduced. Furthermore individuals differ substantially in their ability to correctly classify sentiments. Implementing automated emotion recognition solutions therefore provides a consistent baseline for real-world applications.

1.3. Our work

The goal of this project was to research and understand this technology, followed by the construction, training and evaluation of our own AI model specialized in assessing six basic human emotions: happiness, anger, sadness, fear, disgust and surprise.

The following paper will illustrate the approaches we took to optimize our implementation for accuracy.

2. Related Work

2.1. Efficient Architectures and Low-Resolution Vision

Modern computer vision architectures prioritize efficiency and scalability. **EfficientNetV2** enhances training speed and parameter efficiency through Fused-MBConv layers [25], while **Compact Convolutional Transformers (CCT)** utilize convolutional tokenizers to address information loss on small inputs [11]. Similarly, **ConvNeXt V2** employs masked autoencoders [30], and **Vision Mamba (Vim)** utilizes bidirectional State Space Models for linear complexity scaling [32]. Our work evaluates these architectures, adapting their downsampling strategies for 64×64 facial expression recognition (FER).

2.2. Transfer Learning in Facial Analysis

FER benefits from domain-specific pre-training on massive datasets to overcome data scarcity. Wang et al. [27] demonstrated that an InsightFace ResNet-50 (IR50) backbone, pre-trained on MS-Celeb-1M, captures identity-invariant representations effectively. We adopted these pre-trained IR50 weights to initialize our network.

2.3. Lightweight Models and Transfer Learning

Despite the trend of parameter-heavy CNNs, the **CERN** [9] architecture achieves state-of-the-art performance with only 1.45 million parameters. CERN outperformed larger models, reporting 87.09% and 88.17% accuracies on the in-the-wild RAFDB and FerPlus datasets respectively.

2.4. Optimization and Generalization

Training on heterogeneous datasets requires specialized optimization. A learning rate **warm-up** period stabilizes training with large minibatches by preventing divergence during initial epochs [10]. Furthermore, **Sharpness-Aware Minimization (SAM)** [8] minimizes loss value and sharpness simultaneously, improving generalization on out-of-distribution data.

2.5. FACS-Aligned Benchmark Datasets

The **CK+** dataset provides prototypical expressions annotated via the Facial Action Coding System (FACS) [5], which defines facial movements using Action Units. This FACS-aligned landmark configuration makes CK+ an optimal benchmark for evaluating fine-grained expression dynamics [17].

3. Methodology

3.1. Datasets

To ensure reliable and comparable results, four widely used benchmark datasets from the literature were selected at the beginning of this study. All datasets contain facial images categorized into the six target expressions: anger, disgust, fear, happiness, sadness, and surprise. The datasets employed were a smaller version of AffectNet [1] (22,629 images), CK+ [3] (927 images), FER+ [7] (56,269 images), and RAF-DB [19] (12,135 images).

FER+ was chosen instead of the original FER2013 due to its improved labeling quality, which has a direct impact on model performance and reliability. Moreover, the selected datasets provide diversity in terms of age, ethnicity, and gender, which is essential for improving the generalization capability of the models. Although AffectNet, CK+, and FER+ include an additional *contempt* label, this class was excluded from the experiments as it falls outside the scope of the present study. The data includes both grayscale and RGB images. Image formats vary between JPEG and PNG, which does not affect the preprocessing or training pipeline.

After the initial training sessions, the developed models exhibited strong overfitting. To mitigate this issue, additional datasets were sought, and access requests were submitted to multiple institutions worldwide to expand the training data.

At the end of the data collection process, a total of 310,215 facial images were gathered. In addition to the previously mentioned datasets, the final corpus also included images from EmoSet [26], JAFFE [14], MMA-FEDB [18], ExpWild [6] and WSEFEP [31]. All datasets were carefully curated and harmonized to match the six target emotion categories used in this study.

Several challenges were encountered during dataset collection and curation. Some datasets contained incorrectly labeled samples, including statues, cartoon characters, or animals assigned to human emotion categories. In other cases, images were organized per subject rather than per emotion category, without consistent naming conventions, which prevented automated separation. Furthermore, some small datasets were ultimately discarded due to insufficient sample size or poor labeling quality.

As illustrated in the Figure 1, the aggregated datasets are imbalanced, with a significantly higher number of happiness samples compared to the other emotion categories.

3.2. Cross-Dataset Generalization Strategy

For the final training phase, CK+ and KDEF [15] were deliberately excluded from the training set in order to evaluate the generalization capability of the proposed model on unseen datasets. It is important to emphasize that both datasets are relatively balanced.

KDEF contains high-quality, well-controlled facial images captured under consistent lighting and pose conditions, including photos from multiple angles up to 90º. This increased variation in viewpoint makes it more challenging for our models to accurately predict the expressions, which explains their lower performance on KDEF.

By excluding CK+ and KDEF from training, these datasets served as independent test sets, enabling a more rigorous evaluation of cross-dataset generalization performance.

3.3. Data Pre-Processing Methodology

The pre-processing pipeline functions as a sequential execution of image evaluation modules, where images failing specific quality thresholds are programmatically discarded.

The evolution of this strategy transitioned from strict curation to selective filtering across two empirical phases.

Phase 1: Aggressive Filtering for a Specialized Use Case The primary hypothesis for Phase 1 was that the model would perform optimally if trained exclusively on perfect, high-quality images that mirrored the target demonstration environment (a user looking directly into a high-quality camera). To enforce this pristine standard, the initial pipeline utilized all available filters sequentially:

- **Lighting Filter:** Analyzed the mean pixel intensity and standard deviation, discarding images that were excessively dark, overly bright, or lacked sufficient contrast.
- **Blur Detection:** Assessed image sharpness using frequency analysis via a Fast Fourier Transform (FFT). Images lacking sufficient high-frequency components were rejected.
- **Roll Correction:** Evaluated the image across four rotational states $(0°, 90°, 180°, 270°)$, selecting the orientation that yielded the highest face detection confidence.
- **Yaw Filtering:** RetinaFace detected the face, and 6DRepNet predicted the 3D pose. To guarantee a strict frontal viewpoint, any image with a yaw angle exceeding $45°$ was discarded.
- **Conditional Cropping:** If the ratio of the face area to the total image area was less than 0.4, indicating a small face with significant background, the image was cropped with proportional padding; otherwise, the original image was retained.

In Phase 1, we retained approximately 91,960 images after aggressive filtering.

Phase 2: Resolving the Augmentation Contradiction The aggressively filtered dataset from Phase 1 caused the models to overfit rapidly. To combat this, data augmentation techniques were applied (adding artificial noise, altering brightness, introducing artificial blur, and utilizing MixUp).

This sequence exposed a fundamental contradiction: we were actively removing naturally difficult images during the pre-processing stage, only to artificially recreate those exact conditions during the augmentation stage. Furthermore, augmentations like MixUp produce overlapping visual structures that standard face detectors completely fail to recognize, meaning such images would be universally rejected by strict filters. Recognizing that the model required this data variance to generalize, Phase 2 drastically reduced the aggression of the filtering pipeline:

- **Reduced Aggression:** The universal application of blur, yaw, roll, and lighting filters was halted. For high-quality datasets (AffectNet, CKplus, FERPlus, RAF-DB), the pipeline was stripped down to utilize only structural standardization.
- **Selective Cleaning:** Aggressive techniques were retained only for uncleaned, chaotic datasets (e.g., ExpW and EmoSet), which utilized the cropping module, face detection validation, and manual exclusion of specific emotion labels.

In Phase 2, we started with 310,215 images. After selective pre-processing, we retained 229,478 images.

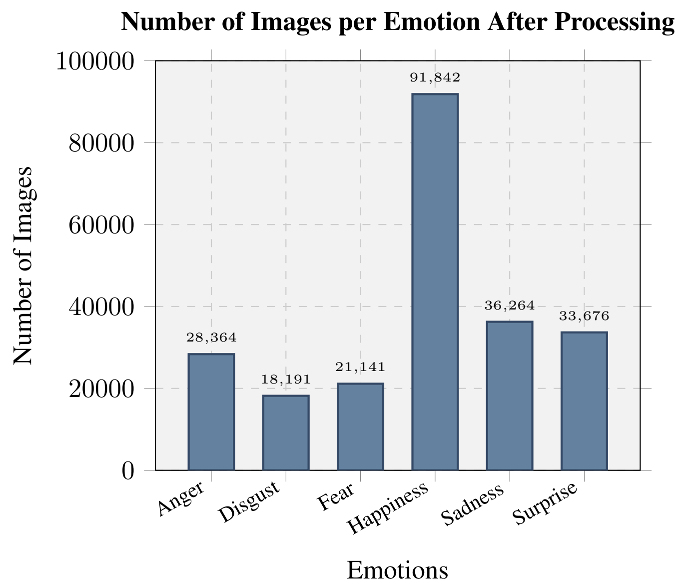

Figure 1. Number of Images per Emotion After Processing

Standardization and Deduplication To finalize the datasets, all images underwent structural standardization:

- **Channel Normalization:** All inputs were converted to a 3-channel RGB format to ensure consistent tensor depth.
- **Resizing:** Images were zero-padded to achieve a square aspect ratio and subsequently resized to exactly 64×64 pixels. We utilized cubic interpolation if the original image was smaller than 64×64.
- **Deduplication:** Duplicate or highly similar images were purged using a combination of perceptual hashing (pHash) and difference hashing (dHash) to ensure strict separation between training and validation distributions.

Dataset Splitting At the beginning, we followed an 80%/10%/10% split strategy, where 80% of the data was used for training, 10% for validation, and 10% for testing. Towards the end of the project, some of us adopted a 90%/10% split strategy in most experiments. The training set comprised 90% of the data, while 10% was reserved for validation (evaluation). For the final test set, we exclusively utilized two specific high-quality datasets: CK+ and KDEF. This rigorous separation ensured that our final performance metrics reflected true generalization on standardized benchmarks.

3.4. Architectural Adaptations

3.4.1 Basic CNNs and ResNet18 models

To ensure experimental comparability, two baseline convolutional neural networks (3-layer and 5-layer CNNs) were implemented alongside a modified version of ResNet-18, in which the initial 7×7 convolution was replaced with a 3×3 kernel and the input resolution was reduced from 224×224 to 64×64.

3.4.2 ResNet18 with Squeeze-and-Excitation

Our model is based on ResNet18 and is extended with a squeeze-and-excitation (SE) attention module. ResNet18 serves as a stable baseline with moderate capacity (roughly 11–12 million parameters). Results from compact FER literature, including CERN, suggest that attention mechanisms can improve performance across different model sizes, supporting our use of SE as a low-overhead architectural enhancement [9].

The input to our network is a standardized RGB face image of size 64x64. Since this is much smaller than the standard ImageNet setting (224x224), we avoid the original ResNet stem (7x7 stride-2 convolution plus stride-2 max pooling), because it would reduce spatial resolution too early and remove fine facial details. Instead, we use a 3x3 convolution with stride 1 and omit early pooling to preserve spatial information [22].

After the stem, we follow the standard ResNet18 design with four residual stages and stage-wise downsampling.

We augment the ResNet18 backbone with squeeze-and-excitation (SE) blocks, which perform lightweight channel-wise attention by learning per-channel scaling factors from globally pooled features, improving representational capacity with minimal overhead [13]. The motivation for using a lightweight attention mechanism is consistent with the goal of improving robustness without greatly increasing model size and is aligned with compact FER approaches such as CERN [9].

3.5. Class Imbalance and Advanced Loss Functions

A significant challenge encountered in our training pipeline is the severe class imbalance within our aggregated dataset, where the *happiness* class constitutes approximately 40% of the total data. Under standard cross-entropy loss, this imbalance introduces a fundamental bias where the majority class becomes the default, high-confidence prediction. This phenomenon is driven by two factors:

- **Frequency versus Difficulty:** The *happiness* class typically presents distinct visual markers (e.g., a smile) that are easily learned. Conversely, minority classes such as *fear* and *disgust* rely on subtle ocular and brow micro-expressions that degrade heavily at lower resolutions, such as 64×64 pixels.
- **Loss Gradients:** Because majority classes occur more frequently, they contribute a disproportionately large share of the gradient updates. The model minimizes the global objective simply by maximizing confidence in the frequent, easy classes, functionally neglecting the minority classes.

To address this imbalance without relying on a `WeightedRandomSampler`, we modified the training objective by evaluating two advanced loss functions:

- **Class-Balanced Focal Loss (CB-FL):** This loss was applied to **both the ResNet18+SE and EfficientNetV2-S models**. It assigns larger weights to underrepresented classes using the effective number of samples [4], while the focal component reduces the contribution of well-classified (easy) samples so that the model continues learning from harder examples [16].
- **Label-Distribution-Aware Margin Loss (LDAM) with Deferred Re-Weighting (DRW):** This loss was evaluated **exclusively on the EfficientNetV2-S model**. LDAM modifies the logit outputs by assigning a significantly larger mathematical margin to minority classes [2]. This strictly penalizes the model unless it maintains extreme confidence when predicting rare classes, thereby enforcing wider decision boundaries around them. The DRW strategy schedules the training such that the model trains without artificial weights for the initial epochs to learn robust feature representations. In the final epochs, strict class frequency weights are applied to fine-tune the decision boundaries for the minority classes without corrupting the initial feature learning phase.

3.5.1 Alternative Architectures

During the evaluation phase, three low-parameter architectures were tested to compare different modern deep learning paradigms. Adapting these models for 64×64 inputs required specific architectural modifications, primarily reducing spatial downsampling.

- **Vim (Vision Mamba, $\sim$7M parameters):** Vim [32] was chosen because it is a novel model challenging standard Transformers by utilizing bidirectional State Space Models (SSMs), promising exceptionally low GPU memory usage. To adapt it for 64×64 images, the patch size was changed from 16×16 to 8×8 to provide a functional sequence length of 64 tokens.
- **CCT-7 (Compact Convolutional Transformer, $\sim$4M parameters):** CCT-7 [11] was selected to test a small Transformer specifically designed to handle small images by using a convolutional tokenizer instead of linear patch embeddings.
- **ConvNeXt V2 Atto ($\sim$3.7M parameters):** This model [30] was chosen to evaluate how a state-of-the-art pure

Convolutional Neural Network operates in a constrained parameter regime. The original architecture features a stem convolution with a stride of 4. For a 64×64 image, this immediately shrinks the feature map to 16×16. The stem stride was explicitly changed to 1 to preserve spatial information.

3.5.2 EfficientNetV2-S Adaptations

EfficientNetV2-S [25] was selected for its MBConv blocks and Squeeze-and-Excitation (SE) attention, which effectively amplify emotional markers like micro-expressions while suppressing background noise.

Architectural and Augmentation Adaptations To process 64×64 images effectively, two major pipeline adjustments were executed:

- **Stem Stride Fix:** The original stem stride of 2 was reduced to 1. This modification preserved the full spatial resolution through the initial layers, preventing the immediate loss of fine facial details.
- **Minimized Destructive Augmentation:** Initial training with standard ImageNet settings (Mixup $\alpha = 0.8$) obliterated facial features. We implemented a low-interference pipeline to minimize data destruction while maintaining regularization. This included reducing Mixup to 0.2, limiting CutMix to a maximum of 0.3, and using gentle affine transformations. This ensured that while the model was forced to reconstruct features, the core emotional signal remained intact.

3.6. Explainable AI

AI systems that only expose the input and output of the model to the user are called ”black box models”. The difficulty this design poses lies in the fact that the developer cannot clearly understand the model’s reasoning process. Explainable AI aims to solve this problem by providing a method to visualize the importance of different image regions to the final decision of the model. For out task we experimented with two different methods:

3.6.1 Grad-CAM

The grad-cam method is widely used to explain computer vision models. The algorithm works by analyzing the gradients used by the last convolutional layer and assigning a level of importance for each neuron’s output for the final classification of the image [20]. However this approach did not seem to work as well as we hoped for our model. Applying grad-cam to our model’s last convolutional layer resulted in a coarse and inaccurate heatmap. To combat this problem we experimented with using the previous convolutional layers as the base layer for grad-cam which yielded strongly improved results as in Figure 2.

Figure 2. Grad-CAM result of the fourth convolutional layer (left), the third convolutional layer (middle), and Integrated Gradients (right)

The poor performance of the grad-cam method is a result of the heavy downsampling during our model’s forward pass. The resolution (8×8) of the final feature maps lacks sufficient spacial information to effectively reconstruct the importance of the low resolution input. This also explains why we achieved better results on a layer using higher resolution feature maps.

3.6.2 Integrated gradients

Integrated gradients is a different algorithm to assess the importance of different image regions for the model’s decision. This method works by attributing importance values to the pixels of a baseline image. This image is slowly transformed into the original input image and in each transformation step the gradients of each pixel are calculated. The Importance of each pixel is then set as the average gradient of each pixel, which represents how much the model’s output changes based on that piece of information [23]. The result is a fine grained attention map which highlights the most important parts of the image as seen in Figure 2.

This approach seemed interesting to us as the quality of the output did not rely on the architectural details of our model.

4. Experiments and Results

4.1. Basic CNNs and ResNet18 Configuration Enhancements

The performance of ReLU and LeakyReLU activation functions was systematically evaluated, with LeakyReLU demonstrating superior overall accuracy. Optimization strategies included Stochastic Gradient Descent (with standard hyperparameters), Adam (with learning rate set as 0.001), and AdamW were tested and Adam with a learning rate of 0.001 yielded the best performance across both the shallow CNN architectures and the modified ResNet-18. Further improvements were achieved through the integration of Batch Normalization, Dropout regularization,

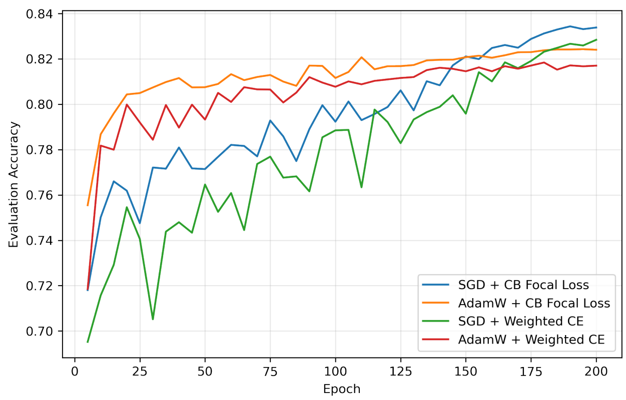

Figure 3. Validation performance comparison for ResNet18+SE under different optimizer and loss configurations.

and late-stage global average pooling, which enhanced feature stability, mitigated overfitting, and preserved high-level spatial representations relevant to emotion classification.

4.2. ResNet18+SE Ablation: Optimizer and Loss Function

To measure the effect of optimizer choice and loss function while keeping the architecture fixed, we trained the same ResNet18+SE model under four configurations: (1) AdamW + class-balanced focal loss, (2) AdamW + weighted cross-entropy, (3) SGD + class-balanced focal loss, and (4) SGD + weighted cross-entropy. All runs used the same input resolution (64×64), the same augmentation setup, and the same training duration. The validation performance curves are shown in Fig. 3.

Across the four settings, SGD with class-balanced focal loss achieved the strongest validation performance. One likely reason is that AdamW can converge quickly early on, but with noisy and heterogeneous FER data it can also lead to solutions that generalize less well. In addition, class-balanced focal loss tends to be more robust for strong imbalance than weighted cross-entropy because it continues to emphasize hard and minority-class examples later in training. In the final pipeline, imbalance is therefore handled mainly through the loss formulation rather than through sampling-based balancing.

4.3. Transfer Learning Evolution (IR50)

To address dataset size limitations, we utilized an InsightFace ResNet-50 (IR50) model, pre-trained on the Ms-Celeb-1M dataset [27]. The adaptation proceeded in three phases. In Version 1, full fine-tuning of the entire architecture led to severe overfitting due to a massive 12.8-million-parameter linear classification head memorizing spatial positions. In Version 2, we froze the initial backbone layers (`input_layer` through `body3`) to preserve pre-trained feature extractors while keeping `body4` trainable. However, the oversized head still caused a training-validation accuracy gap exceeding 10 percent. Ultimately, Version 3 replaced the massive head with a 795,000-parameter "Light Head" and incorporated a Spatial Attention module (Conv1x1 with Sigmoid activation) before Global Average Pooling to retain local facial features. With reduced trainable parameters (5.8 million), modified MixUp, and Exponential Moving Average (EMA) decay, this final architecture prevented overfitting and achieved a validation accuracy of 81.38 percent.

4.4. Alternative Architectures Evaluation

4.4.1 ResNet18+SE vs. ResNet18+CBAM

To evaluate whether a stronger attention mechanism improves cross-dataset generalization, we replaced the squeeze-and-excitation (SE) blocks in our ResNet18 backbone with CBAM [29] and compared both variants under the same training protocol. In both cases, the models were trained on the aggregated training corpus while explicitly excluding CK+ and KDEF. These two datasets were then used as independent test sets to measure generalization to unseen, well-controlled benchmark distributions.

Overall, both attention variants performed similarly in the sense that they achieved reasonable accuracy on the held-out datasets, but SE consistently yielded higher test accuracy. ResNet18+SE reached 84.69% on CK+ and 67.33% on KDEF, whereas ResNet18+CBAM achieved 81.97% on CK+ and 62.67% on KDEF. On the global held-out evaluation set ("All"), ResNet18+SE achieved 68.47% compared to 63.94% for ResNet18+CBAM. These results suggest that the additional spatial attention component in CBAM did not translate into improved robustness for our low-resolution (64×64) FER setting, while the simpler channel-only SE mechanism provided slightly more stable generalization across datasets.

4.4.2 Vim (Vision Mamba):

However, Vim is fundamentally designed for high-resolution data, and it proved too difficult to make the architecture focus effectively on the fine details of small facial images.

4.4.3 CCT-7 (Compact Convolutional Transformer):

Even though the architecture was perfectly suited for small images, it simply possessed too few parameters to learn the complex features required for human emotion classification.

4.4.4 ConvNeXt V2 Atto:

Similar to CCT-7, the model lacked the parameter count necessary to build robust representations.

4.4.5 MobileNet V2

With ≈ 3.5M parameters MobileNet V2 is one of the smallest Networks we implemented. Sadly the low parameter count and adaptation to images of size 64×64 were incompatible with out goal of effective emotion recognition. The model only achieved an accuracy of 66% on the evaluation data and was hence discarded from future experiments.

4.4.6 Other ResNet implementations

Regarding the high accuracy of the ResNet18 + SE implementation ResNet34 and ResNet50 were also trained and evaluated. However the training process was less efficient since the models started overfitting early on due to the high parameter count. As a result the model evaluation yielded a lower accuracy than the original ResNet18 + SE design.

4.5. EfficientNetV2-S Optimization Dynamics

EfficientNetV2-S ultimately emerged as one of the top two best-performing models across all validation benchmarks. To ensure a rigorous comparison, we empirically optimized and tuned each optimizer to find its stable configuration.

Experiment 1: SGD with Nesterov Momentum

- **Result:** Peaked at 74.25 percent validation accuracy.
- **Performance:** SGD [10] proved highly sensitive to hyperparameters. Slight alterations in the learning rate caused massive performance swings, ranging from a complete failure to learn (stagnating below 40 percent accuracy) to achieving a moderate peak of approximately 75 percent.
- **Why it underperformed:** Even at optimal settings, SGD struggled with structural incompatibilities. It applies a uniform learning rate across the network, causing it to under-train depthwise layers while over-training pointwise layers. It lacks the variance tracking required to stabilize the contradictory signals arising from 10 distinct data sources, and it struggles to converge when targets are ambiguous soft labels from Mixup.

Experiment 2: AdamW

- **Result:** Peaked at 83.10 percent validation accuracy.
- **Why it succeeded:** AdamW's adaptive per-parameter learning rates successfully navigated the heterogeneous architecture, balancing updates between depthwise and dense layers. It provided consistent convergence but exhibited mild overfitting in the final epochs.

Experiment 3: SAM + AdamW (Sharpness-Aware Minimization)

- **Result:** Peaked at 82.46 percent validation accuracy.
- **Performance Dynamics:** SAM [8] improves generalization by minimizing the worst-case loss in the immediate neighborhood of the parameters, forcing the model into flat, stable minima. While it slightly underperformed the standard AdamW on the validation dataset, it demonstrated superior generalization stability. The drop in accuracy between the validation and test datasets was smaller for SAM, indicating robust out-of-distribution performance. However, its final absolute accuracy on the test dataset remained nearly identical to the simple AdamW baseline, differing by approximately -1 percent. Ultimately, this marginal improvement in generalization stability was outweighed by the severe computational burden of evaluating the network twice per step.

4.6. Evaluation of Class Imbalance and Prediction Bias

Evaluating the EfficientNetV2-S baseline model exposed a severe prediction bias: despite a global accuracy of 60.17%, the network over-predicted *happiness* (96.62%) while failing on *fear* (17.20%) and *disgust* (29.99%). To address this disparity, we evaluated two advanced anti-bias functions: Class-Balanced Focal Loss (CB-FL) and Label-Distribution-Aware Margin Loss (LDAM) with Deferred Re-Weighting (DRW).

While these techniques successfully forced the network to fixate on difficult images, performance improvements were strictly limited to the training dataset. During true cross-dataset testing on unseen CK+ and KDEF faces, minority class accuracies remained stagnant or declined, and global accuracy fell by approximately 3%. This performance degradation stems from three primary factors:

- **Dataset Memorization:** Chasing strict mathematical fairness on small, noisy distributions caused the models to overfit to specific training lighting and identities rather than generalizable features.
- **Data Ambiguity:** Subjective emotions like *fear* and *disgust* contain inherent label noise. Forcing the algorithm to perfectly separate these overlapping concepts exacerbates overfitting.
- **Hyperparameter Tuning:** The degradation might stem from suboptimal configurations. For example, setting margin boundaries or focal penalties too aggressively can severely disrupt generalization, indicating a need for further systematic tuning to discover the optimal balance.

4.7. GAN

Inspired by the work with Super-Resolution Generative Adversarial Network [21] we decided to test with a 2x upscaling factor applied at our 64x64 images and then resize it back to 64x64.

Furthermore, a Super Resolution Generative Adversarial Network (SRGAN) method is a GAN that can convert low-resolution images into high-resolution and realistic images. [28]

Unfortunately, due to the image size and that the enhancement process makes the images a bit softer with less facial marks, the training with GAN-enhanced images was worse than with the normal images, achieving easily more than 15% difference between train and evaluation accuracy.

The chosen GAN was GFPGAN, a practical algorithm for real-world face restoration. The model with the second version had the best experimental result and was selected to be applied to all the images, so the group could compare between the normal folder and the GAN-enhanced folder.

4.8. Dual-Input Training

Inspired by the paper about CERN [9] the dual-input strategy was implemented. It consists of giving the training process the same image as two inputs and the correct label. The first image is the normal one and the second is augmented. So in order to receive the correct result, the model has to predict both versions of the image correctly. The results with the dual-input were outperformed by the normal training. It also has increased overfitting, a difference bigger than 10% (86.45% to 76.15%) between train and evaluation accuracy.

4.9. CERN-Based Architecture

The implemented CERN architecture is a computationally reduced LightCNN-based model incorporating MFM activations, Efficient Channel Attention, and a dual-input consistency training strategy adapted to hardware constraints. At epoch 47, the model achieved 66.70% evaluation accuracy and 67.51% training accuracy. The learning dynamics showed extended plateaus followed by sudden improvements, suggesting that additional training time could potentially yield further performance gains. However, limited computational capacity—including the use of an NVIDIA RTX 2060 GPU—prevented full exploration of longer training schedules or the use of pretrained weights from the original seven-class model.

5. Conclusion and Retrospective

This project demonstrates that targeted architectural choices and rigorous evaluation methodologies profoundly impact the success of Facial Expression Recognition systems. The ResNet18+SE architecture performed the best overall, outperforming alternative model by approximately 2 percent. Furthermore, the evaluation of newer architectures, such as the Vision Mamba (Vim) network, revealed that they generalize exceptionally well relative to their minimal parameter counts, confirming that the latest model architectures generally outperform older legacy networks.

Because emotion datasets are typically small, leveraging transfer learning is highly beneficial for establishing ro-

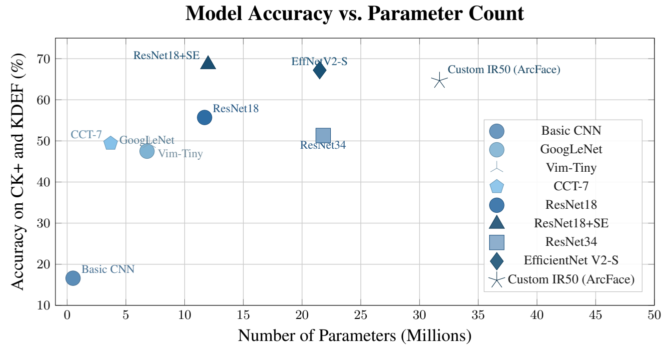

Figure 4. Model Accuracy vs. Parameter Count evaluated on the CK+ and KDEF datasets.

bust initial feature extraction. Extensive data augmentation also proved incredibly helpful for improving model generalization. To accurately measure this generalization, we established that reserving complete, separate datasets exclusively for testing is the ideal evaluation methodology. The training dynamics confirmed that optimal performance requires pairing specific architectures with distinct optimizers, as no single optimizer excelled across both EfficientNet and ResNet configurations.

Attempts to resolve prediction bias utilizing advanced mathematical loss functions, such as Class-Balanced Focal Loss, were ultimately not significantly more successful. These anti-bias measures often induced dataset memorization rather than solving the underlying data ambiguity. The fear emotion remained hard to guess. Additionally, when attempting to interpret the networks, Grad-CAM was not precise enough for small input images; Integrated Gradients proved to be a significantly better option for extracting reliable, high-resolution visual explanations.

5.1. Retrospective

Future iterations of this research should continue to prioritize the search for the latest architectural developments and aggressively leverage transfer learning from massive facial datasets. There is also a distinct need to allocate more computational resources toward systematically finetuning all hyperparameters. Most importantly, further research must address the severe prediction bias toward the *happiness* class. Subsequent work must investigate exactly why complex, subtle emotions are predicted so poorly and develop targeted, data-driven strategies to improve minority class recognition without compromising global accuracy.

A. Extended Description of the CERN Architecture Implementation

The CERN architecture is derived from a LightCNN backbone with depths ranging from 9 to 29 layers in its full configuration. LightCNN is designed to provide compact yet discriminative feature representations, particularly for face-related tasks. It employs Max-Feature-Map (MFM) activations, which perform competitive feature selection by suppressing weaker feature responses through channel-wise max operations. This mechanism reduces redundancy, improves generalization, and lowers computational overhead compared to conventional activation functions.

To enhance representational capacity without significantly increasing complexity, the architecture integrates Efficient Channel Attention (ECA). Unlike heavier attention mechanisms, ECA performs lightweight channel-wise recalibration using local cross-channel interaction without dimensionality reduction. This allows the model to emphasize informative feature channels while maintaining computational efficiency.

A central component of the framework is its dual-input training strategy. During training, the network processes both an original image and its augmented counterpart simultaneously. The prediction is considered correct only if label consistency is maintained across both inputs. This constraint encourages the network to learn perturbation-invariant and robust feature representations, improving generalization under variations such as illumination changes, noise, and geometric transformations.

Due to GPU memory constraints on the university server, it was not feasible to replicate the complete architecture including the combined local and global attention modules described in the original formulation. Therefore, a reduced variant consisting of seven convolution layers was implemented. The batch size was limited to 16 samples to remain within available memory limits.

Despite these simplifications, the model remained computationally demanding. Training for 10 epochs required more than 24 hours of runtime. Extended training sessions were occasionally interrupted due to system constraints, making it necessary to implement checkpoint-based continuation to preserve training progress and ensure experimental reproducibility.

B. Appendix A: Detailed Architecture of the Light Head (Version 3)

The final InsightFace ResNet-50 (IR50) architecture utilized in this study replaces the standard classification head with a custom "Light Head" to mitigate severe overfitting.

Spatial Attention and Dimensionality Reduction: The original IR50 classification head flattened the spatial grid into a 25,088-dimensional vector, feeding it into a 12.8-million-parameter linear layer. This caused the network to memorize exact spatial positions rather than generalized emotional features. To resolve this, we utilized a Global Average Pooling (GAP) layer. Because human emotions are highly localized (e.g., specific micro-expressions around the eyes or mouth), a standard GAP layer would destroy critical spatial context. Therefore, we introduced a Spatial Attention module immediately preceding the GAP layer. This module consists of a Conv1x1 layer followed by a Sigmoid activation, allowing the network to assign learnable weights to important spatial areas before dimensionality reduction.

Parameter Efficiency and Training Metrics: The implementation of the Light Head reduced the parameter count of the classification head from 12.8 million to 795,000, bringing the total trainable parameters of the model down to 5.8 million (including the unfrozen `body4` block). To stabilize training further and eliminate the memorization effect, we reduced the MixUp probability and applied an Exponential Moving Average (EMA) decay. This configuration successfully closed the previously observed 10 percent gap between training and validation accuracy, achieving a peak validation accuracy of 81.38 percent.

C. Contributions

Dmytro

Model Training and Experimentation: Led the training and evaluation of EfficientNetV2-S (and its associated configurations), CCT-7, Vim-tiny, and explored the ConvNeXt V2 architecture. Conducted transfer learning experiments utilizing the IR50 backbone. Executed the final robustness tests for the ResNet18+SE model across various hyperparameters, which confirmed baseline stability despite yielding identical final accuracies.

Data Pipeline and Preprocessing: Enhanced and developed segments of the data preprocessing pipeline by implementing Fast Fourier Transform (FFT) for an improved version of blur filtering, introducing facial roll correction, yaw correction, and optimizing boundary margin resizing to ensure minimal residual margins post-cropping or square image outputs. Collaborated directly with Tiago to determine and execute the optimal combination of preprocessing operations tailored to each specific dataset.

System Integration and Explainability: Engineered a live demonstration script designed to process video input and output continuously annotated video streams, integrating real-time emotion classification alongside Explainable AI (XAI) visual overlays.

Research and Organization: Sourced approximately 90 percent of the datasets introduced during the second project phase. Conducted extensive literature reviews to identify and propose alternative network architectures. Facilitated project management by structuring team meetings and designed the foundational architecture of the project's README documentation.

Aleksandar

Lighting filter : A preprocessing module for removing poorly exposed images

A `LightingFilter` was developed to analyze intensity statistics and discard samples that are excessively dark, overly bright, or lack sufficient contrast. The filter was used during the early preprocessing phase to improve overall data quality and reduce noisy training samples.

RGB standardization : A preprocessing utility for consistent input format

A `toRGB` conversion step was added to standardize all inputs to a consistent $H \times W \times 3$ RGB format. This ensured compatibility across datasets containing mixed grayscale and RGB images and simplified downstream tensor processing.

ResNet18 + SE model : Final backbone and attention design

A ResNet18 backbone with Squeeze-and-Excitation (SE) attention blocks was implemented and became the final architecture used for the best-performing configuration. A ResNet18+CBAM variant was also trained and evaluated; however, results were very similar and SE performed slightly better, so the SE-based architecture was retained.

Imbalance-aware loss functions : Loss design and evaluation

Class-balanced focal loss was introduced as the primary strategy for handling class imbalance, alongside weighted cross-entropy as a baseline. Empirical comparisons showed that class-balanced focal loss achieved stronger performance, leading to its adoption in the final pipeline.

Optimizer and loss ablations : Training experiments and comparison plots

Controlled ablation studies were conducted for ResNet18+SE to compare optimizer and loss combinations, including SGD vs. AdamW and class-balanced focal loss vs. weighted cross-entropy. The resulting training curves and performance comparisons were included in the final report.

Explainable AI prototypes : Development and evaluation of interpretability methods

A Grad-CAM prototype was created as an initial interpretability baseline, but produced coarse explanations at 64×64 resolution. Additional attribution methods were evaluated, including occlusion-based analysis and Integrated Gradients. Integrated Gradients yielded the most informative and fine-grained explanations and was used in the final qualitative analysis.

Additional dataset contribution : Dataset discovery and preprocessing

The DFEW dataset was identified as an additional data source and contributed to the group. Preprocessing and dataset-specific cleaning steps were created to integrate DFEW into the standardized pipeline.

Tiago

- Conducted literature research on CNNs, ResNet architectures, GANs, data augmentation techniques, and face detection methods to define the project's technical direction.
- Performed dataset investigation and detailed quality analysis, including data distribution, class balance, image consistency, and label reliability. Compared original datasets with GAN-augmented datasets to evaluate performance impact.
- Implemented baseline CNN models with 3 and 5 convolution layers and adapted ResNet-18 to the project scope. Conducted extensive experiments involving batch normalization, variations in average pooling, dropout tuning, data augmentation strategies, dual-input training setups, and comparative evaluation using GAN-generated images.
- Adapted and optimized the GAN implementation to align with project requirements and ensure compatibility with the university server environment.
- Developed preprocessing pipelines including resizing and cropping using MTCNN, later replacing it with RetinaFace, and analyzed their impact on model performance.

Daniel

Dupicate finder : A class that finds duplicate images

This class can be configured the pHash or dHash algorithm for fuzzy hashing. It exposes a method called `find_duplicates` expects a list of filepaths as an input. The given files are then loaded in batches, hashed and stored in a Dictionary if the hamming distance is higher than a given threshold. The Dictionary is used to avoid unnecessary comparisons (images are in the same "Bucket if the first 32 Bits match"). The method returns a list of paths to duplicte images for further processing.

Blur filter : A class that measures the blurriness of a given image

This class is part of the group's preprocessing pipeline. Given an image the class computes the laplacian variance of the input and removes it if the variance is lower than a given threshold. This implementation was later adapted to use the *fast fourier transform* (FFT) to classify the image.

Different Model implementations : Different custom implementations that were trained

During the development phase of the project I wanted to evaluate as many different model architectures as possible. Due to the fixed input size of (64×64) all models were implemented from scratch and modified to downsample less aggressively. The developed models were all trained using a custom training script.

Model Evaluation script : A script that given a Model implementation measures its accuracy

This script was later used by all members of the group to measure and compare their model's performance on unseen datasets.

Image demo : A script that processes all images in a directory

This script expects the path to a directory filled with images, loads all images in batches, loads our model and evaluates all images. The resulting probabilites for each emotion are then written into a csv file with the related filepath

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